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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
III Semester B.E. April -2024 Examinations
DEPARTMENT OF MATHEMATICS
LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS
(Common to EC, EE, EI, ET)
(2022 SCHEME)
MODEL QUESTION PAPER

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A

1	1.1	Consider the polynomials $p_1(t) = 1 + t^2$, $p_2(t) = 1 - t^2$. Is $\{p_1, p_2\}$ a linearly independent set in P_3? Justify your answer.	2
	1.2	Write which matrix reflects every vector in $\mathbb{R}^2$ about the line $y = x$. Also find the transformation of the vector $(-3,\ 2)$?	2
	1.3	Show that the set $\{u_1, u_2, u_3\}$ is an orthogonal basis of $\mathbb{R}^3$, where $u_1 = (3,-3,0)$, $u_2 = (2,2,-1)$, $u_3 = (1,1,4)$.	2
	1.4	Choose the second row of $A = \begin{bmatrix} 0 & 1 \\ a & b \end{bmatrix}$ so that A has eigen values 4 and 7.	2
	1.5	In Fourier series expansion of f(x) = x2 in the interval -2 < x < 2 the value of an is _____.	2
	1.6	Find Fourier transform of a single ramp given by $f(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases}$.	2
	1.7	For a certain data $\sum x^2 = 2156$, $\sum y^2 = 2822$ and $\sum xy = 1784$ then find the co-efficient of correlation for the data.	2
	1.8	The Fourier series coefficient a_0 for the signal $x(t) = e^{-2t}$, $0 \le t \le 2$ is_____.	1
	1.9	At the point of discontinuity, Fourier series of $f(x) = \begin{matrix} -\pi, -\pi < x < 0 \\ x,\ 0 < x < \pi \end{matrix}$ converges to _____.	1
	1.10	If the Fourier transform of $e^{-\frac{x^2}{2}}$ is $\sqrt{2\pi}\, e^{-\frac{s^2}{2}}$, then the Fourier transform of e^{-2x^2} is_____.	1
	1.11	The Fourier cosine transform of x^3 is $\frac{4}{s^4}$, then Fourier sine transform of $x^3 sin3x$ is _____.	1
	1.12	If $\beta_2 = 0.3$, then comment on the Kurtosis of the distribution.	1
	1.13	If between x and y variables has no correlation the angle between lines of regression is _____.	1

PART-B

		UNIT-I	
2	a	Show that the set of all vectors of the form $(2s + 4t, 2s, 2s - 3t, 5t)$ is a subspace of R^4.	4

	b	Obtain the bases for the column space and left null space of the matrix $A = \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}$.	6
	c	Find the linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^4$such that $T(e_1) = (0,1,0,2), T(e_2) = (0,1,1,0), T(e_3) = (0,1,-1,4)$. Also find the rank and nullity of the linear transformation.	6

		UNIT-II	
3	a	Using the Gram-Schmidt process compute the QR factorization of the matrix $A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$.	8
	b	Find the SVD of the matrix $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$.	8
		OR	
4	a	Determine least square solution for the inconsistent system $AX = B$, where $A = \begin{bmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -2 \\ 0 \\ 8 \end{bmatrix}$.	8
	b	Obtain the matrix P such that it diagonalises the matrix $A = \begin{bmatrix} 7 & -4 & 4 \\ -4 & 5 & 0 \\ 4 & 0 & 9 \end{bmatrix}$. Also find the inverse of P.	8

		UNIT-III	
5	a	Express f(x) = e^{-ax} in the interval (-π, π) as a Fourier Series and hence show that $\frac{\pi}{\sin n\pi} = 2 \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2+1}$.	8
	b	The voltage applied to the primary winding of a transformer is a rectangular wave function defined by $f(x) = \begin{cases} 0, & -2 < x < -1 \\ k, & -1 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$. Find the Fourier series representation.	8
		OR	
6	a	Obtain the complex Fourier series for $f(x) = \cos(ax)$, where 'a' is non-integer real constant, over the interval (0, 2π). Deduce the corresponding trigonometric Fourier series.	8

b. The following values of y give the displacement of a certain machine part for the rotation x of the flywheel: (8)

x	0^0	60^0	120^0	180^0	240^0	300^0	360^0
y	1.98	2.15	2.77	-0.22	-0.31	1.43	1.98

Express y in Fourier series up to the second harmonic.

UNIT-IV

7 a. Find Fourier transform of a single pulse of amplitude v_0 given by $f(t) = \begin{cases} v_0, & |t| < 1 \\ 0, & |t| > 1 \end{cases}$. Hence evaluate the integral $\int_0^\infty \frac{\sin t}{t} dt$. (8)

b. Obtain Fourier Sine transform of the function $\frac{x}{x^2+1}$. Hence prove that $\int_0^\infty \frac{t^2}{(t^2+1)^2} dt = \frac{\pi}{4}$. (8)

OR

8 a. Find Fourier transform of a parabolic pulse given by $f(t) = \begin{cases} 1-t^2, & |t| < 1 \\ 0, & |t| > 1 \end{cases}$ Hence evaluate the integral $\int_0^\infty \frac{t\cos t - \sin t}{t} \cos\frac{t}{2} dt$. (8)

b. Use Parseval's identity, prove that $\int_0^\infty \frac{dx}{(a^2+x^2)(a^2+y^2)} = \frac{\pi}{2ab(a+b)}$. (8)

UNIT-V

9 a. Calculate the first four moments of the following distribution about the mean, also find β_1, β_2 and comment upon the skewness and kurtosis of the distribution. (10)

Variable	1	2	3	4	5	6	7	8	9
Frequency	1	6	13	25	30	22	9	5	2

b. The following table gives the scores obtained by 11 students in English and Kannada translation. Find the rank correlation coefficient. (6)

Scores in English	40	46	54	60	70	80	82	85	85	90	95
Scores in Kannada	45	45	50	43	40	75	55	72	65	42	70

OR

10 a. The following table gives the stopping distance y in meters of a motor bike moving at a speed of x $Kms/hour$ when the breaks are applied. (8)

x	16	24	32	40	48	56
y	0.39	0.75	1.23	1.91	2.77	3.81

Find the correlation coefficient between the speed and stopping distance and estimate the maximum speed at which the motor bike could be driven if the stopping distance is not to exceed 5 meters.

	b	Determine the equation of the regression plane connecting x_1, x_2 *and* y. Estimate y at $x_1 = 1.8$, $x_2 = 112$.	8

Diffusion time (hour) x_1	1.5	2.5	0.5	2.6	0.7	1.6
Sheet-resistance (Ohms-cm) x_2	66	87	69	93	66	123
Current gain y	5.3	7.8	7.4	10.8	6.5	12.6

Signature of Scrutinizer:
Name:

Signature of Chairman
Name: